FORWARD DEPLOYED ENGINEERING

LEC-03

1. Executive Summary

- **Extensible Architecture:** AI workers leverage vendor-free standards like **Agent Skills** for procedural knowledge and **Model Context Protocol (MCP)** for software integration.
- **Scheduled Tasks & Resource Management:** Automated tasks are configured with specific frequencies, times, and delivery methods, balanced against strict platform resource limits and cost-value priorities.
- **Stateful Isolation via Projects & Memory:** AI context and data are securely isolated using projects, custom instructions, and selective general/project-specific memory mechanisms.

2. Core Concepts & Definitions

Term / Concept	Definition / Explanation
Agent Skills	Vendor-free industry standards allowing AI agents to gain procedural knowledge on how to perform tasks.
MCP	An industry standard connector framework allowing AI agents to safely read and write data with real-world software.
Context / RAG	Uploaded documents and files referenced via semantic search so the AI can retrieve relevant information when needed.
Forward Deployment Engineer (FDE)	A technical role bridging the gap between AI development and enterprise implementation using platforms and SDKs.

3. Structured Body Hierarchy

- **Extending AI Worker Capabilities**
    - **Skills & Connectors:** Agents learn procedural tasks via Agent Skills and connect to external software via Model Context Protocol (MCP), ensuring vendor-free interoperability.
    - **LLM Statefulness:** While raw foundational LLMs are stateless, surrounding platform infrastructure (memory, context, environment configuration) makes them stateful.
- **Scheduling, Prioritization & Resource Limits**
    - **Task Configuration:** Define clear intent, requirements, frequency (daily/weekly), execution time, and delivery channels (push notification, session, email).
    - **Cost vs. Value:** Prioritize automated tasks by weighing their utility against token consumption and resource costs.
    - **Usage Limits:** Platforms like Cloude enforce strict usage thresholds (e.g., 5-hour session caps) to manage resource consumption.
- **AI Memory, Projects & Context Management**
    - **Memory Control:** Segmented into general, project-specific, topics, people, and projects, with options to disable creation or exclude sensitive topics.
    - **Projects & Instructions:** Project-specific memories and custom instructions guide agent behavior and isolate workflows. Instructions are always included in prompts, while memories are selectively recalled.
    - **Context / RAG:** Uploaded files act as searchable reference libraries rather than active prompt injections.

4. Review & Active Recall (Q&A)

- **Q: What is the difference between Agent Skills and MCP connectors?**
    - **A:** Agent Skills provide procedural knowledge (*how* to perform a task), while Model Context Protocol (MCP) connectors enable real-world software integration to read and write data.
- **Q: How do custom instructions differ from project memories?**
    - **A:** Custom instructions are persistently sent with every prompt to govern behavior, whereas project memories are selectively recalled by the AI as needed.
- **Q: Why must scheduled tasks be prioritized based on value and cost?**
    - **A:** To optimize token consumption and prevent exceeding strict platform resource limits (such as hourly session caps and weekly quotas).
- **Q: How does a project-specific memory differ from general memory?**
    - **A:** Project-specific memories are strictly isolated to that specific project container and workflow, while general memories apply across broader, cross-project interactions.